Hindi Verbal Fluency: VFT + SpAM Mental Lexicon Analysis

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INTRODUCTION AND DATASET

- This study explores how Hindi speakers search their mental lexicon using two tasks: Verbal Fluency Task (VFT) and Spatial Arrangement Method (SpAM).
- 35 Hindi-English bilingual participants
- VFT: 60-sec word generation in animals, foods, colours, body-parts
- SpAM: participants arranged words by similarity in 2D.

PRIOR WORK

- Phonological similarity influences semantic memory retrieval showing that sound structure can bias access to meaning [1].
- Words that are phonologically similar tend to also be semantically related more than expected by chance [2].
- Spatial Arrangement Method efficiently captures human semantic similarity judgements by arranging items based on perceived relatedness [3].

RESEARCH QUESTIONS (RQ1-RQ7)

- **RQ1**: Do semantic categories differ in productivity (words) or retrieval speed (IRT)? - **H1, H2, H3, H4**
- **RQ2**: Does participant spatial structure align with semantic or phonetic models? - **H5, H6**
- **RQ3**: Does an integrated VFT+SpAM score outperform VFT-only? - **H9**
- **RQ4**: Does phonological-over-semantic similarity increase over retrieval order? - **H7**
- **RQ5**: Does phonological similarity predict lexical productivity? - **H8**

METHODOLOGY

- Analysis was limited mainly to Hindi and Hinglish responses from the four target domains: animals, body-parts, colours, and foods.
- VFT data was used to study word productivity, retrieval time, and semantic clustering.
- Statistical testing included Welch's t-test, Kruskal-Wallis tests, and linear mixed-effects models for repeated participant data.
- Semantic and phonetic patterns were measured using the multilingual sentence-transformer model google/muril-base-cased
- Semantic embeddings were built from the original words, while phonetic embeddings were built from transliterated phonetic keys.
- SpAM cluster layouts were compared with semantic and phonetic cluster structures using ARI and NMI.
- An integrated VFT+SpAM score was also compared with the VFT-only score.

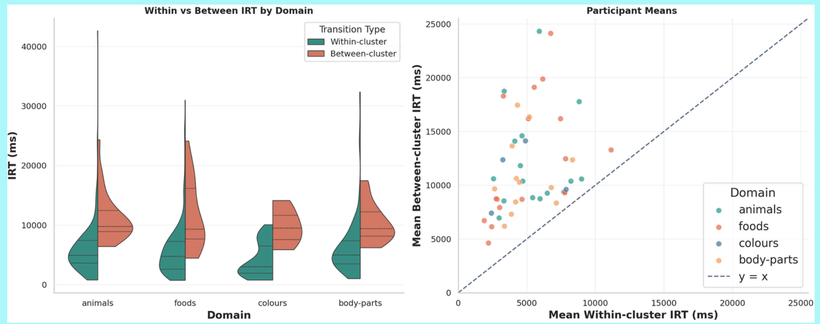
HYPOTHESES TABLE (H1-H9)

Hypothesis (description)	Test used	Result	Signif.	Interpretation
H1: Between-cluster IRT > within-cluster IRT (semantic boundary cost)	Welch t	t=9.30, d=1.12	p<0.001	Supported strong effect.
H2: Positive serial slowing (lexical exhaustion)	Linear Mixed Effect Models	No positive slope (0/4 domains)	p>0.05	Not supported.
H3: Domain effect on total words	Kruskal	H=3.264	p=0.196	No domain difference.
H4: Domain effect on IRT	Kruskal	H=3.903	p=0.142	No domain difference.
H5: Semantic ARI above chance	Sign + Perm.	68/86 positive; mean ARI=0.1765	sign<0.001, perm=0.502	Mixed/inconclusive.
H6: Phonetic ARI above chance	Sign + Perm.	52/86 positive; mean ARI=0.0745	sign=0.066, perm=0.503	Not supported.
H7: Phon-over-sem similarity increases with position	Mixed model	beta=+0.003127	p=0.0905	Not supported.
H8: Higher phonological similarity -> higher productivity	Spearman	rho=+0.114	p(one)=0.257	Not supported.
H9: Integrated vs VFT-only correlation differs	Steiger	z=0.000; rho=-0.461 both	p=1.000	No difference.

SEMANTIC VS PHONETIC SUMMARY

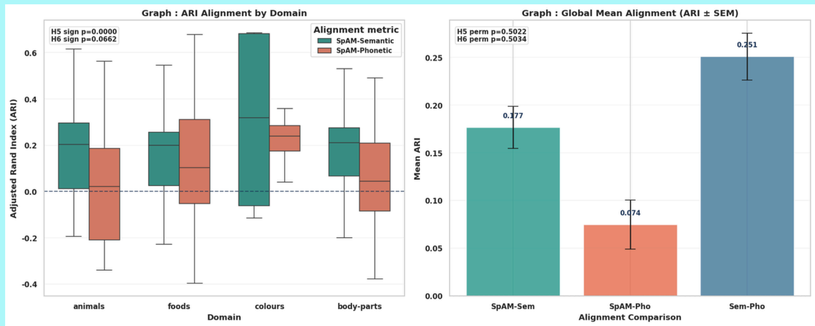
Semantic similarity: weak-to-moderate structure present (H5 mixed).
Phonetic similarity: no clear confirmatory support (H6/H8 not supported).

BETWEEN-CLUSTER IRT > WITHIN-CLUSTER IRT (PLOT)



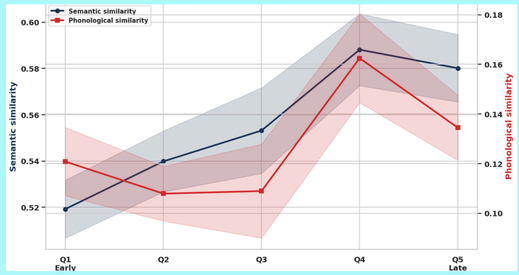
Interpretation: between-cluster pause is clearly higher than within-cluster pause.

CLUSTER ALIGNMENT



Interpretation: semantic ARI tends to be above phonetic ARI, but effect is modest.

SEMANTIC VS PHONOLOGICAL SIMILARITY ACROSS RETRIEVAL ORDER



Interpretation: no reliable phonological facilitation trend over retrieval order.

CONTRIBUTIONS

- Hypothesis-Based Split: H1,H4,H7 (Akshat) | H2,H5,H8 (Om) | H3,H6,H9 (Ankit)
  - Each member gets 3 hypotheses spanning foundational to advanced, ensuring equal difficulty progression.
  - Interleaved Collaboration: Simultaneous code + analysis + writing across all hypotheses with shared frameworks (permutation tests, clustering); weekly integration checkpoints.

DISCUSSION AND CONCLUSION

- Hindi speakers do not seem to search words randomly; retrieval is structured and shows a strong semantic boundary cost.
- VFT results suggest that words are produced in meaningful clusters, while SpAM results show that participant layouts reflect some semantic and phonetic organization.
- Domain-level differences in word output and retrieval time were not strong, and phonological effects were weak or inconsistent.
- The combined VFT+SpAM score did not clearly outperform the simpler VFT-only score.

Hindi speakers appear to search their mental lexicons through organized semantic neighborhoods, with some additional phonetic structure, rather than through random retrieval.

LIMITATION

- Dataset is too small, even in that, so many English words, which resulted in more lesser data.
- Small and uneven sample across domains (especially colours) limits statistical power and generalizability.

REFERENCES

[1] Kumar et al. (2022) - Phonology in semantic memory retrieval
[2] Dautriche et al. (2016) - Wordform & semantic similarity
[3] Hout et al. (2013) - Spatial Arrangement Method for semantic similarity

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